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RESEARCH ARTICLE

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Numerical investigation on the influence of mesoscopic deformation on contact area and hydraulic aperture to fracture seepage

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Abstract

Fluid flow through a single fracture is commonly described by the cubic law. However, deviations from this model are expected because natural fracture surfaces are rough and in contact with each other in discrete regions. In this study, the interactions between fracture closure, contact area, and hydraulic characterization of mesoscopic-scale rough fractures were investigated. First, natural-splitting fractures induced by Brazilian splitting were generated and digitally reproduced using a three-dimensional scanner. The distribution characterization of the initial mechanical fracture apertures was described, and the variation in fracture contact area during fracture closure was evaluated. Second, numerical simulations of fluid flow between rough surfaces were conducted, and the Stokes equations were solved. The results reveal that the hydraulic aperture underwent three stages. When the contact area was less than 0.34%, changes in the hydraulic aperture resulted in flow rate changes that were consistent with the cubic law. When the contact area was between 0.34% and 14.41%, the hydraulic aperture gradually deviated from the cubic law and fell below the mean fracture aperture. At this stage, the hydraulic aperture should be fully considered rather than the mean fracture aperture. When the contact area exceeds 14.41%, although the mechanical fracture decreases with increasing strain, the influence of the contact area on the hydraulic aperture gradually diminishes. This is because the fractures contain less fluid, and the hydraulic aperture approaches the minimum. Finally, the relationship between mechanical and hydraulic aperture strains was established, improving geological engineering applications that account for fracture deformation.

KEYWORDS

contact area, hydraulic aperture, mesoscopic deformation, rough fracture, seepage characteristics

Highlights

- Variation in the contact area of a mesoscopic fracture during fracture closure.
- Fluid flow in a mesoscopic Fracture with fracture closure.
- Relationship between flow and strain in a mesoscopic fracture.

1 | INTRODUCTION

Fractures are significant drivers of fluid flow in subsurface rocks with low matrix permeability. Although the hydraulic properties of fractures have been extensively investigated from numerical and analytical perspectives, the accurate representation and prediction of natural fractures remain more challenging (Lei et al., 2015; Panja et al., 2022; Velasco et al., 2018). This challenge stems from the multitude of random rough fractures and their diverse stress conditions (Bandara et al., 2021; Dou et al., 2019; Feng et al., 2022; Li et al., 2023). Consequently, exploring the influence of mesoscopic deformation on the seepage of random rough fractures is imperative for devising a more pragmatic approach for characterizing hydraulic properties. Such an approach is crucial for the safety and performance evaluation of various engineering geological endeavors, including hydrogeological surveys, geological disposal of nuclear waste, petroleum reservoir exploration, carbon dioxide geological storage, and groundwater contamination prevention (Bonneau et al., 2022; Han et al., 2021; Pan et al., 2023; Zou, 2023; Zhang et al., 2023).

The cubic law is a common model used to simulate the flow in a single fracture. In most applications, the fluid flow through a single fracture is assumed to be analogous to the laminar flow between two perfectly smooth parallel plates; the fluid is viscous and incompressible, and flow rate Q is proportional to the square of fracture aperture e:

$$Q = -w\frac{e^3}{12\mu}\frac{\mathrm{d}p}{\mathrm{d}x}, \tag{1}$$

where e is fracture aperture, which refers to the vertical distance between two smooth platen; μ is viscosity; p is seepage pressure lampressure; and w is the fracture width in the direction normal to fluid flow. If we compare Equation (1) to Darcy's Law, the intrinsic permeability of the fracture is $k = e^2/12$ (Chen, Zhang, et al., 2023; Xing et al., 2021).

However, owing to the presence of unsmooth elements, local aperture variations, and asperities, the fracture walls are rough, and geometric contact occurs at discrete points under in situ stress (Lavoine et al., 2024; Lei et al., 2021; Niu et al., 2022; Zhang et al., 2022). Numerous experimental and numerical studies have shown that fluid flow in rough fractures tends to follow paths of least resistance, bypass contact areas, and flow in open voids, deviating from the cubic law (Chen et al., 2017; Develi & Babadagli, 2015; Wei et al., 2023b). Therefore, to better describe the fluid seepage characteristics in a rough fracture, several definitions of the "aperture" were proposed. First, a mechanical aperture e_{m} was suggested (Brown et al., 1985), and local mechanical apertures distribution $e_{\mathrm{m}}(x, y)$ was considered the local distance between the surfaces. Second, another relevant definition is mean aperture $<e>$ (Brown, 1987). It can be expressed as follows:

$$\langle e \rangle = \frac{1}{L_x L_y}\int_{x=0}^{L_x}\int_{y=0}^{L_y} e(x, y)\mathrm{d}x\mathrm{d}y, \tag{2}$$

where L_x and L_y denote the fracture dimensions. Mean aperture $<e>$ more closely represents the aperture available to flow. Subsequently, hydraulic aperture e_{h} was developed to evaluate the actual transmissivity of a rough fracture (Chen, Zhang, et al., 2023; de Dreuzy et al., 2001; Gao et al., 2023; Xiong et al., 2011). According to e_{h}, the transmissivity of a rough fracture can be correlated with its geometry by varying the aperture and obstructed regions. However, it is challenging to obtain an equivalent hydraulic aperture for natural rough fractures. Therefore, various modifying coefficients, such as the fractal dimension, joint roughness coefficient, and standard deviation of e_{m}, have been developed to modify hydraulic aperture (Chen et al., 2017; Han et al., 2021; Walsh et al., 1981; Zimmerman et al., 1996). Various prediction models for hydraulic apertures have been proposed using various methods and techniques in laboratory experiments, such as digital image correlation, three-dimensional (3D) laser scanning (Develi & Babadagli, 2015; Guo et al., 2022; Kim et al., 2022; Xiong et al., 2018), and numerical simulations (Kashefi et al., 2021; Wang et al., 2022; Wei et al., 2023b). In addition to the fracture aperture, the fracture closure under normal stress is important for fracture seepage. Several detailed studies have been conducted on the influence of contact on fracture seepage (Chen et al., 2015; Matsuki et al., 2008; Renshaw, 1995; Wei et al., 2023a).

Parameters such as the fractal dimension and roughness are difficult to obtain in on-site testing. Moreover, the fracture deformation with in situ stress in a deep rock mass varies even in the same fracture. Therefore, it is crucial to establish a relationship between laboratory and onsite testing to obtain the hydraulic characteristics of fractured rock masses. In recent years, strain measurement technology has undergone rapid development, and strain gauges, laser Doppler vibrometry, and fiber optical monitoring have been widely used in mining, tunnels, slopes, and other engineering projects (Chai et al., 2024; Liu et al., 2022; Wang et al., 2024). The high-strain accuracy of the microstrain order ensured its effectiveness. Several researchers have proposed using strain to initiate research on hydraulic characteristics, and a linear trend between the mechanical aperture strain and hydraulic aperture strain has been reported (Fidelibus, 2007; Liang et al., 2025; Ma et al., 2023).

The preceding discussion highlights that fracture deformation under normal stress is characterized by the fracture strain and fractional contact area, both of which exert definitive control over the fluid flow. Thus, further research on the influence of mesoscopic deformation on the contact area and hydraulic aperture of fracture seepage was conducted to investigate the following: (1) the influence of the mesoscopic deformation of rough fractures on the contact area and mechanical aperture, (2) the interdependence between the contact area and fluid flow behavior, and (3) the relationship between the fluid flow and strain in a single mesoscopic fracture under normal stress. The remainder of this paper is organized as follows: Section 2 introduces the methods used for sample preparation, fracture surface morphology analysis, fracture closure calculation, and numerical simulation of fluid flow. Section 3 analyzes the measured and simulated results of mesoscopic deformation on the contact area and the hydraulic characterization of fracture seepage. Finally, the conclusions are presented in Section 4.

2 | METHODOLOGY

2.1 | Sample preparation

To investigate the impact of mesoscopic fracture deformation on the contact area and hydraulic aperture, it is ideal to utilize fractures induced by hydraulic fracturing in situ. However, obtaining such fractured specimens is challenging owing to complex geological conditions. Fractures formed using the Brazilian splitting or shear method are similar to the mechanical causes of natural fractures, and the resulting fracture surface morphology is more consistent with reality (Goodman et al., 1976; Zhang et al., 2023, 2025). In this study, the Brazilian splitting method was used to generate the rough fractured surfaces. The procedure for creating the fractures is as follows. First, natural granite was cut into standard cylinders with dimensions of Φ50 mm × 100 mm, and both ends of each cylinder were polished by a lapping machine. Subsequently, the specimens were placed in splitting molds, as shown in Figure 1. Finally, as the normal displacement of the splitting wedge increased, the specimens fractured, yielding two fracture surfaces measuring 50 mm × 100 mm.

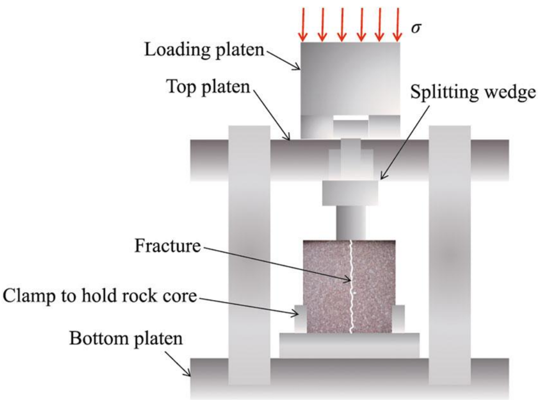

FIGURE 1 Generation process of splitting fractures with a Brazilian splitting device.

2.2 | Fracture surface morphology

To accurately characterize the geometry of fractures and explore their effect on fluid flow, it is essential to first obtain the 3D geometric morphology data of rough fractures. In this study, the fracture surface topography was measured using a 3D laser scanner (EinScan Pro2X, Konica Minolta Investment Ltd.). This scanner can offer 20 μm precision in the x, y, and z coordinates, high measurement speed, and favorable repeatability. After scanning, the topographic data were imported into the modeling software, 3D Studio Max, to recreate the 3D rough surface of each fracture. Figure 2 shows the 3D surface topography of each sample. The color transition reflects the height variation of the rough fracture surface. The fractures are labeled Frac1–Frac8, and FracX-1 and FracX-2 denote the upper and lower wall surfaces of the rough fractures, respectively.

2.3 | Fracture aperture

Fracture aperture is a crucial factor that influences the hydraulic behavior and is typically positively correlated with it. After modeling, a 3D fracture surface topography data set was obtained. However, the data set was unordered, resulting in the height data on the upper and lower surfaces not being aligned. To characterize the fracture aperture between the surfaces, the process of obtaining the distribution of the fracture aperture is as

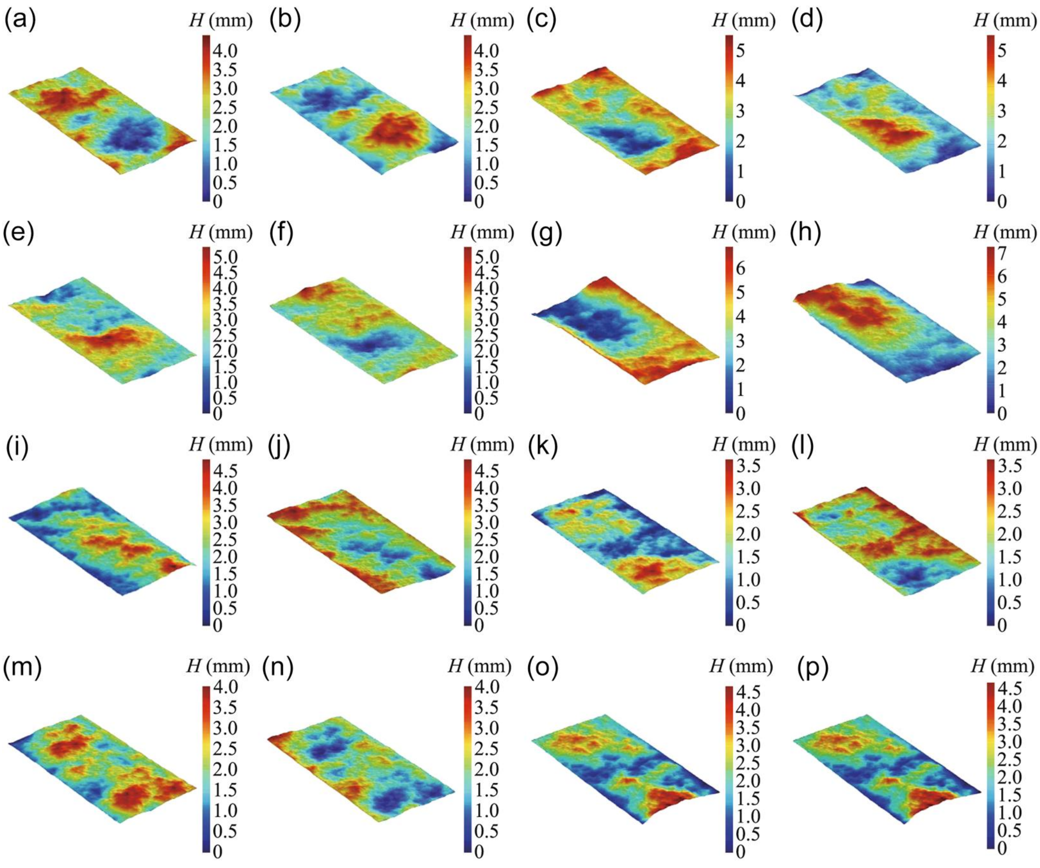

FIGURE 2 3D digital surface topography of each sample: (a) Frac1-1, (b) Frac1-2, (c) Frac2-1, (d) Frac2-2, (e) Frac3-1, (f) Frac3-2, (g) Frac4-1, (h) Frac4-2, (i) Frac5-1, (j) Frac5-2, (k) Frac6-1, (l) Frac6-2, (m) Frac7-1, (n) Frac7-2, (o) Frac8-1, and (p) Frac8-2.

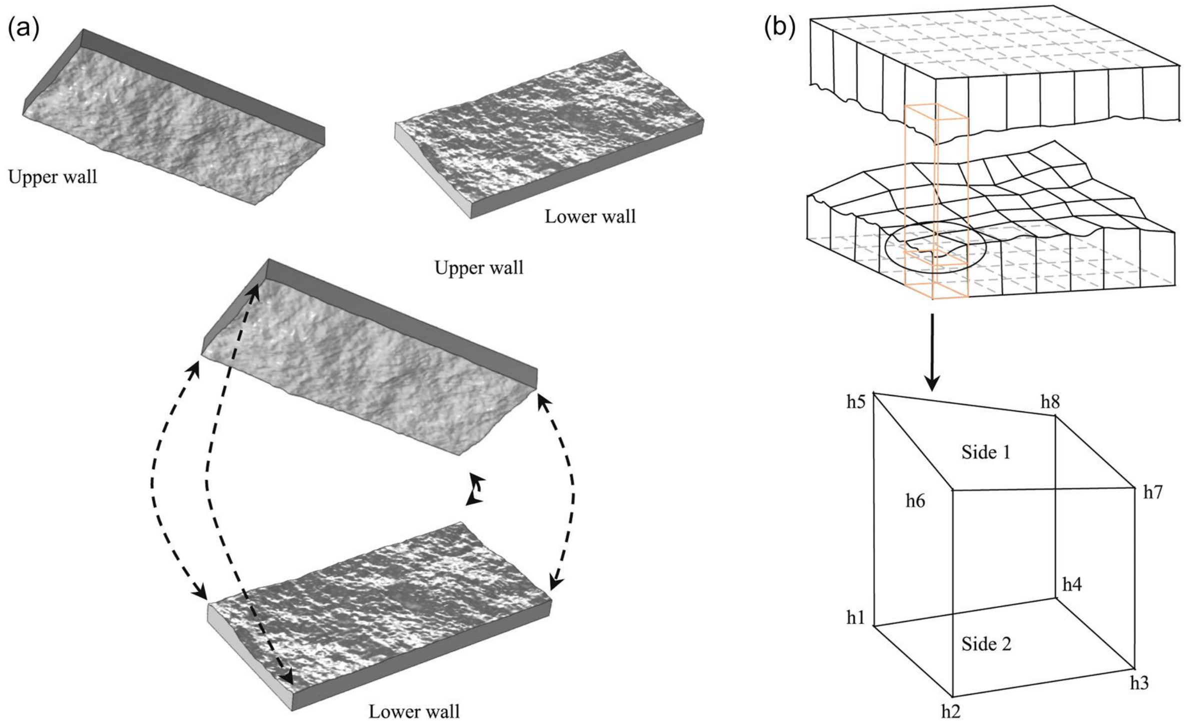

FIGURE 3 Calculation steps for fracture aperture. (a) Schematic diagram of fracture alignment method, (b) schematic diagram of initial fracture aperture calculation.

follows, as depicted in Figure 3. First, the 3D fracture surface topography data set was loaded into Sufer 15.0 software. The Kriging interpolation method was used to divide the fracture region into regular cells in the spatial range in the x- and y-directions with a node spacing of 0.02 mm. Thus, the error introduced by the division is limited, as illustrated in Figure 3. Subsequently, the fracture alignment method proposed by Tatone et al. (2012) was employed to splice the upper and lower fracture surfaces based on their respective morphologies and typical feature points. In this manner, the initial local mechanical fracture aperture, $e_{m0}(x,\ y)$, was obtained, and the initial mean fracture aperture was calculated using Equation (2). In this study, the initial fracture aperture distribution mainly pertained to fractures in the natural state, that is, under zero normal stress conditions.

2.4 | Fracture closure

As the normal stress increases, the upper and lower walls of the fracture move closer to each other. Deformed mechanical fracture aperture $e_m(x,\ y)$ is identical to the difference between initial mechanical fracture aperture $e_{m0}(x,\ y)$ and normal displacement u, which can be written as follows:

$$\begin{cases} e_m(x, y) = e_{m0}(x, y) - u & e_{m0}(x, y) > u, \\ e_m(x, y) = 0 & e_{m0}(x, y) \le u. \end{cases} \tag{3}$$

When $e_m(x,\ y)$ is less than or equal to the normal displacement, the geometric contact of the fracture walls occurs at the corresponding $(x,\ y)$ coordinate. Many analytical and empirical models have been developed to describe the relationship between the normal stress and fracture closure, considering the initial normal stiffness, elastic modulus, and maximum mechanical fracture aperture, such as the Goodman and Bandis models. This study focused on the geometric contact caused by fracture closure. To better evaluate the influence, mechanical fracture aperture strain ε_m was used to describe the closure of a single rough fracture (Ma et al., 2023), which can be expressed as follows:

$$\varepsilon_m = \frac{e_{m0} - e_m}{e_{m0}} = 1 - \frac{e_m}{e_{m0}}. \tag{4}$$

2.5 | Fluid flow

For realistic fracture geometries in which the walls are neither smooth nor parallel, the full Navier–Stokes (N–S) equations are very hard to solve, either analytically or numerically (Zimmerman & Paluszny, 2024). Consequently, various approximations are made to reduce the N–S equations to simpler equations that are more tractable and analyze flow in rough fractures.

Stokes' equations are one of the various approximations of the N–S equations. It is widely recognized and applied in the numerical simulation of fluid flow within a fracture. It neglected the nonlinear effects inherent in the N–S equations and can solve the flow of fluid in three-dimensional rough fractures more conveniently. Therefore, in this study, the Stokes equations for incompressible and steady Newtonian fluids are used and can be expressed as follows:

$$\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} + \frac{\partial^2 v_x}{\partial z^2} = \frac{1}{\mu}\frac{\partial p}{\partial x}, \tag{5}$$

$$\frac{\partial^2 v_y}{\partial x^2} + \frac{\partial^2 v_y}{\partial y^2} + \frac{\partial^2 v_y}{\partial z^2} = \frac{1}{\mu}\frac{\partial p}{\partial y}, \tag{6}$$

$$\frac{\partial^2 v_z}{\partial x^2} + \frac{\partial^2 v_z}{\partial y^2} + \frac{\partial^2 v_z}{\partial z^2} = \frac{1}{\mu}\frac{\partial p}{\partial z}, \tag{7}$$

where v_x, v_y, and v_z are the x, y, and z components of the flow rate, respectively.

The built-in modules of the COMSOL Multiphysics software were used to solve the continuity and Stokes equations for the fluid flow. For the finite element method, the meshing of the physical field is a key step that directly determines the accuracy of the numerical simulation results and the solution time. In COMSOL, mesh delineation can be performed by selecting the existing mesh options; the finer the mesh is, the more accurate the numerical simulation results. However, a finer mesh significantly increases the computation time. A preliminary simulation calculation shows that there is a numerical difference between the conventional, refined, and relatively fine hydrodynamic data. However, there is almost no difference in the data between ultrafine and extremely fine grids, though ultrafine grids greatly save computation time compared with extremely fine grids. Therefore, to balance the accuracy and computational cost, we chose an ultrafine grid to delineate the physical field.

Figure 4 shows the computational domain and boundary conditions of the numerical simulation. The computational domain was a 50 mm × 100 mm rectangle. The inlet was defined as a constant injection pressure of 10 Pa, and the outlet was defined as an open boundary. The direction of the flow was along the length, and the sidewalls were set as no-slip boundaries. The coupling effect between the fluid flow and the rock was not considered. Boundary probes 1 and 2 were placed at both ends of the rough fracture model to monitor the velocity and flow rate at the entrance and exit, respectively. Based on Equation (4), the normal displacement of the fracture under different strains can be obtained. Subsequently, the fracture aperture under different strains loaded into the software can be obtained. This study conducted a hydraulic characteristic analysis of Frac1–Frac8 under 24 conditions ($\varepsilon_m = 0$, 0.025, 0.050, 0.075, 0.100, 0.125, 0.150, 0.175, 0.200, 0.250, 0.300, 0.350, 0.400, 0.450, 0.500, 0.550, 0.600, 0.650, 0.700, 0.750, 0.800, 0.850, 0.900, and 0.950).

3 | RESULTS AND DISCUSSION

3.1 | Distribution characterization of initial mechanical fracture apertures

Understanding the distribution characteristics of fracture apertures is crucial for investigating fracture mesoscopic deformation and seepage. To analyze the distribution of the initial mechanical fracture apertures, the frequency histograms of the data set of $e_{m0}(x, y)$ of Frac1–Frac8 were plotted. In Figure 5, "Max" indicates the maximum vertical distance between the upper and lower walls of the fracture, max $e_{m0}(x, y)$, "Std" denotes the standard deviation of fracture apertures, and "Mean" represents mean fracture aperture $<e>$. The Green curve shows the cumulative percentage of the fracture apertures. According to fracture closure rules, contact exists between the closest points of the upper and lower walls, resulting in a minimum aperture of 0 mm for each fracture.

Numerous studies (Chen, Che, et al., 2023; Ma et al., 2023) have demonstrated that aperture distribution can be described using a normal distribution function. Figure 5 illustrates the distribution of the fracture apertures for Frac1–Frac8. The normal distribution function effectively captured the characteristics of the fracture apertures. However, because of the inherent randomness of the fractures, the mean, standard deviation, and skewness of each fracture exhibited variability. The mean, standard deviation, and maximum aperture ranged from 0.2406 to 1.0277 mm, 0.0587 to 0.1601 mm, and 1.1428 to 2.4414 mm, respectively. The detailed parameters for Frac1–Frac8 are presented in Table 1.

3.2 | Variation in the contact area of a mesoscopic fracture during fracture closure

With an increase in the normal strain, the upper and lower walls of the fracture came closer together. When the normal displacement of node u exceeds or equals the aperture of any node, a geometric contact occurs between the upper and lower walls of the fracture at that node. The fractional contact area, denoted by c, is defined as the ratio of fracture contact area S_{con} to fracture projection area S_{frac}. S_{con} can be obtained by integrating the region in which e_m is zero. Based on the definition of $c = S_{frac}/S_{con}$, when $c = 0$, the upper and

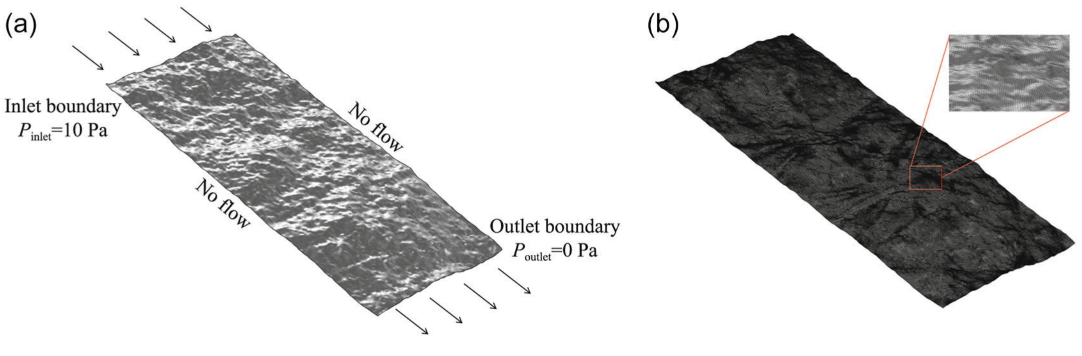

FIGURE 4 Boundary settings of the numerical simulation. (a) Boundary conditions, (b) mesh refinement model.

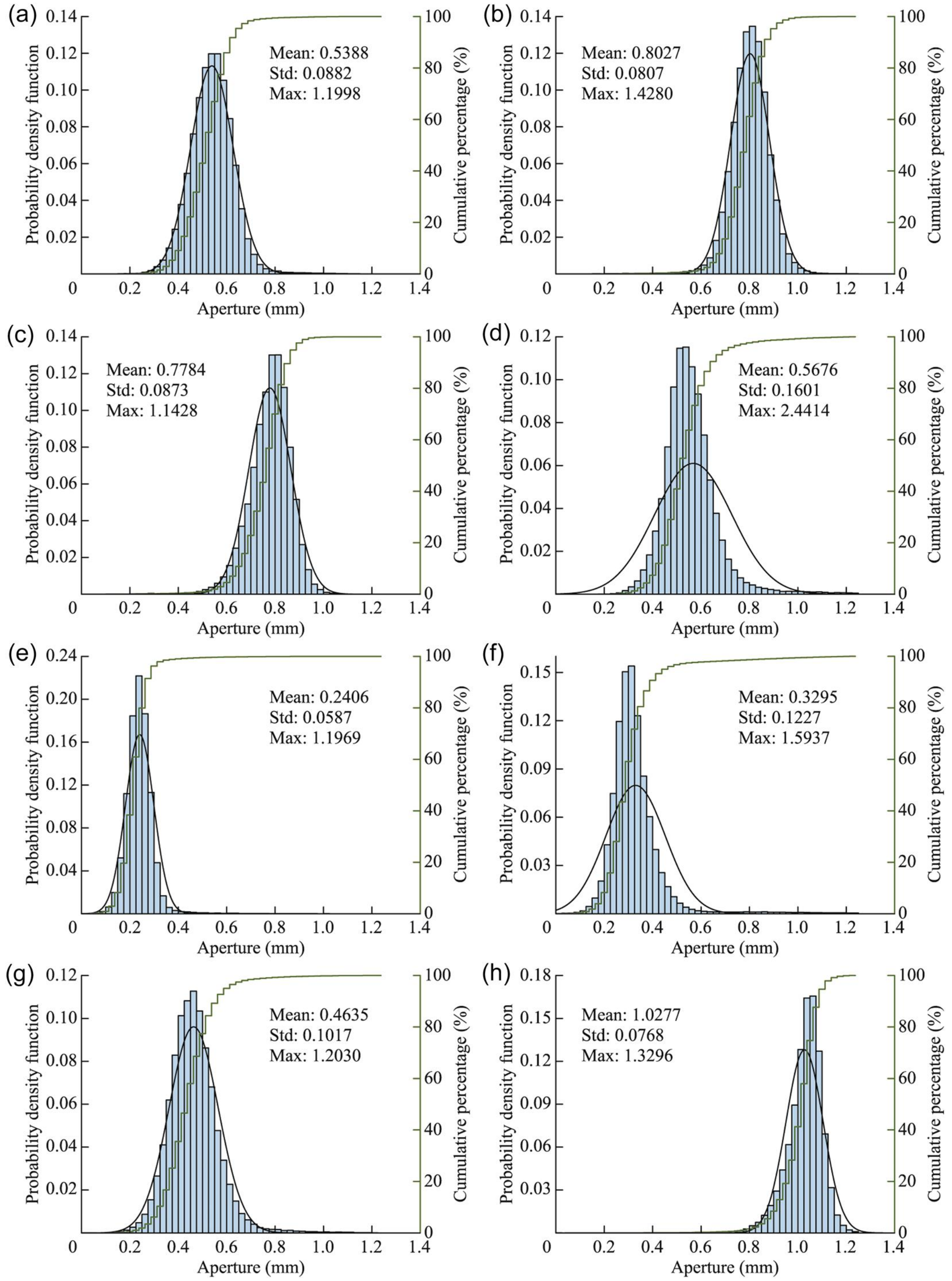

FIGURE 5 Histogram of frequency distribution of different micro fracture apertures. (a) Frac1, (b) Frac2, (c) Frac3, (d) Frac4, (e) Frac5, (f) Frac6, (g) Frac7, and (h) Frac8.

TABLE 1 Specific parameters of fracture aperture distribution of Frac1–Frac8. mm

Sample No.	Mean	Std	Max	Sample No.	Mean	Std	Max
Frac1	0.5388	0.0882	1.1998	Frac5	0.2406	0.0587	1.1969
Frac2	0.8027	0.0807	1.4280	Frac6	0.3295	0.1227	1.5937
Frac3	0.7784	0.0873	1.1428	Frac7	0.4635	0.1017	1.2030
Frac4	0.5676	0.1601	2.4414	Frac8	1.0277	0.0768	1.3296

lower walls of the fracture are not in contact with each other, indicating that the fracture is completely open. For $0 < c < 1$, the upper and lower walls of the fracture were in contact and partially closed, respectively. Moreover, c increases with an increase in normal stress and fracture closure. When $c = 1$, the fracture is completely closed (Table 2).

The distribution of the contact area during fracture closure determines the fluid flow path and behavior. Figure 6 shows the distributions of the fracture closure and opening areas under different strains of Frac-1. With increasing strain, the c value of Frac-1 changed significantly. Corresponding to the three stages of c mentioned above, it can be seen that (1) when $\varepsilon < 0.3$, the fracture walls are mainly open areas, and a small number of areas are closed. The c value remains unchanged. (2) When $0.3 \leq \varepsilon \leq 0.6$, c experiences a significant increase from 1.97% to 98.3%. (3) When $\varepsilon > 0.6$, c is close to 0, indicating that the fracture is closed.

Specifically, Figure 7 shows the probability density function for the fracture aperture of Frac1 under different strain conditions. As the fracture closed, significant changes occurred in the distribution of fracture apertures. The distribution of the fracture aperture underwent an overall leftward movement. Initially, areas with small apertures were the first to close. Because of the low density of this part, its effect on c was minimal; however, the mean fracture aperture significantly decreased from 0.5388 to 0.2983 mm. Subsequently, as the walls moved closer, c gradually increased, and the areas with the fracture opening were more concentrated. When $\varepsilon = 0.45$, the aperture with the highest frequency in the fracture begins to close, and the corresponding c value increases to 50.53%, with more than half of the areas closed. This suggests that c is determined by the initial mechanical fracture aperture distribution. The contact rate was approximately equal to the cumulative percentage of the fracture aperture corresponding to the closure. Therefore, it is important to better understand the specific closure of fractures. When $\varepsilon = 0.6$, although the mean is 0.0922 mm and the max aperture is 0.4799, most of the fracture is closed, and the opening apertures are distributed discretely within the fractures.

Figure 8 illustrates the change in c from Frac1–Frac8 with the strain. Despite the fracture aperture following a normal distribution function, changes in c vary with the strain. For example, the c values of Frac5 and Frac6 begin to change at approximately $\varepsilon = 0.1$, and the fracture is essentially closed at $\varepsilon = 0.25$, indicating that fracture contact and closure occur from 0.1 to 0.25. The change in c occurs at $\varepsilon = 0.6$ for Frac8, with the fracture closing at $\varepsilon = 0.9$. This range was significantly larger than those of Frac5 and Frac6, and the strain corresponding to its entry into the change stage occurred significantly later than those of Frac5 and Frac6. A detailed analysis of the initial fracture aperture distribution corresponding to these fractures revealed that although the maximum and average fracture apertures for Frac5 and Frac6 were different, they exhibited similarities under strain conditions (the same ratio of deformation displacement to maximum fracture aperture). This suggests that for different rough fractures, c is influenced by the aperture distribution, and the change in the contact rate is determined by the mean and maximum values of the apertures.

3.3 | Fluid flow in a mesoscopic fracture with fracture closure

The c value of the fracture affects the fluid behavior and path; however, this does not imply that c is linearly related to the hydraulic aperture or permeability of the fracture. This is because the flow paths of the fluid varied with the fracture closure. Once the fluid flow paths closed, the permeability of the fractures rapidly decreased. Zimmerman et al. (1993) demonstrated that the location of the contact area does not play an important role in the permeability until $c > 0.5$.

To illustrate the relationship between contact area and fluid flow in a mesoscopic fracture, Figure 9 displays the standardized seepage velocity field $u/u_{\max}$ of Frac1, along with flow rate Q of the outlet and corresponding contact area c at different strains ε. When $\varepsilon \leq 0.2$, although Q decreased from 6.21×10^{-8} to 1.00×10^{-8} m^3/s, the

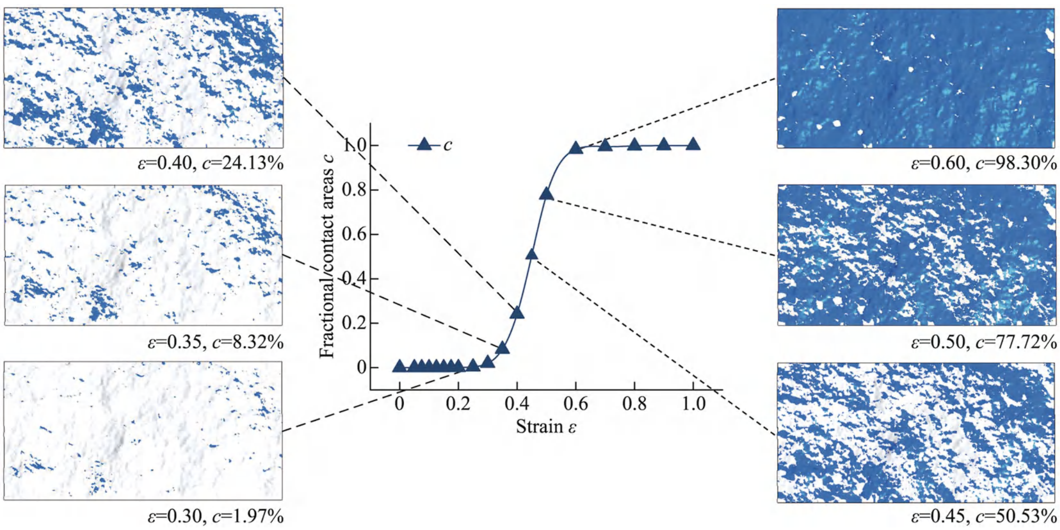

FIGURE 6 Distribution of fracture closure and opening under different strains of Frac1.

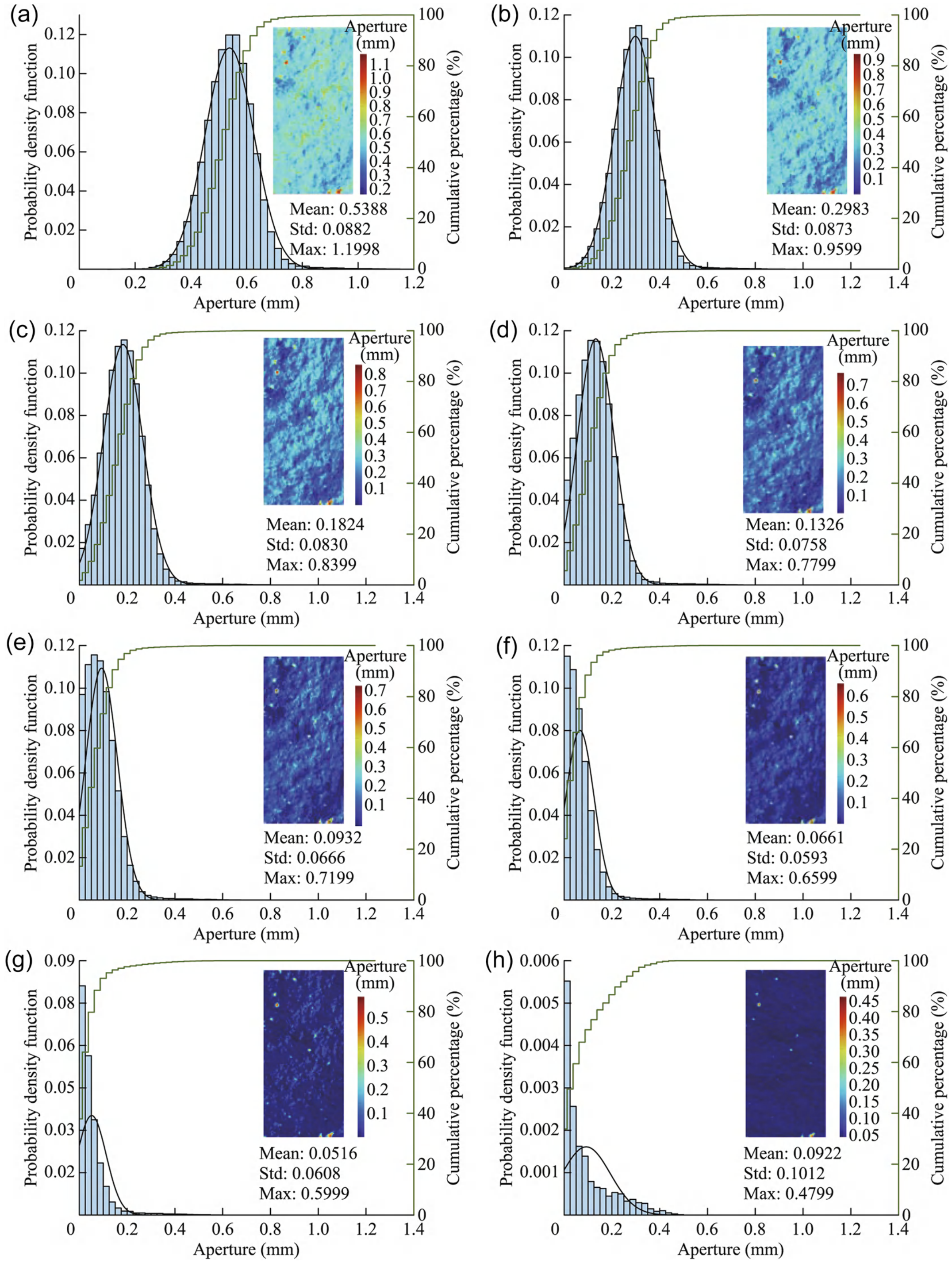

FIGURE 7 Probability density function for fracture aperture of Frac-1 under different strain conditions. (a) $\varepsilon = 0$, (b) $\varepsilon = 0.20$, (c) $\varepsilon = 0.30$, (d) $\varepsilon = 0.35$, (e) $\varepsilon = 0.40$, (f) $\varepsilon = 0.45$, (g) $\varepsilon = 0.50$, (h) $\varepsilon = 0.60$.

standardized seepage velocity field and fluid flow path remain relatively constant. This suggests that, at this stage, fracture closure does not significantly alter the flow path. The decrease in Q was primarily influenced by the fracture aperture, and the impact of c could be disregarded. In subsequent stages, from $0.2 < \varepsilon < 0.4$, there is a nonlinear reduction in Q from 1.00×10^{-8} to 6.76×10^{-11} m^3/s, whereas the corresponding c value increased from 0.36% to 24.13%. During this stage, the fracture closure significantly affects the fracture seepage, reducing not only the fracture aperture but also altering the fluid flow path. The standardized seepage velocity field exhibits preferential flow, transitioning from evident to diminished. When $0.4 < \varepsilon < 0.6$, c rapidly increased from 24.13% to 93.66%, with a corresponding Q decrease from 6.46×10^{-11} to 5.28×10^{-16} m^3/s. The standardized seepage velocity field exhibited a dark region, indicating the disappearance of the preferential flow. This suggests that despite c being only 24.13%, the fracture seepage was weak. This is significantly lower than the criterion of $c < 0.5$.

According to the cubic law, the hydraulic aperture of a smooth fracture is equivalent to the mean fracture aperture, that is, the vertical distance between two smooth plates. Similarly, when applying the cubic law to simulate the seepage between two noncontact rough fracture surfaces, the hydraulic aperture remains equal to the mean fracture aperture. This point has been proven in many previous studies and in the analysis in this study. However, this linear relationship may be altered when contact occurs between the walls of the rough fractures. To more effectively display the influence of fracture deformation and closure in the hydraulic fracture aperture, the relationship between R, c, Q, and strain ε was plotted in Figure 10.

Here, R is defined as the ratio of e_h to $<e>$. When $R \approx 1$, it indicates that e_h is identical to $<e>$, suggesting that the contact area has a minor impact on hydraulic aperture e_h. $<e>$ can predict e_h well. When $0 < R < 1$, it signifies that e_h is less than $<e>$, implying that the contact area weakens e_h. When $R \approx 0$, it indicates that the hydraulic aperture is significantly less than the average fracture aperture. As shown in Figure 4, the seepage process was divided into three stages. Taking Figure 10a as an example, first, when $R \approx 1$ (A value greater than 0.95 is accepted), the c value has a slight increase from 0% to 0.36%, whereas the flow rate of outlet Q significantly decreases from 6.2×10^{-8} to 4.95×10^{-9} m^3/s with increasing ε. This indicates that in this stage, e_h is mainly influenced by $<e>$, and c has a weak impact on e_h. Second, when $0.95 > R > 0.50$, c rapidly increased from 0.36% to 93.66%, and Q decreased slowly from 4.95×10^{-9} to 5.28×10^{-16} m^3/s. At this stage, the increase in c significantly weakened e_h, with R noticeably decreasing from 0.96 to 0.02. In other words, e_h changed from being initially equal to e_m to being significantly smaller. The contact areas close the flow paths and alter the seepage velocity. Third, when $R < 0.05$, c increased slowly from 93.66% to 99.99%, and most areas were closed. Although $<e>$ linearly decreases with increasing ε, Q remains practically constant at 4.09×10^{-16} m^3/s. This indicates that e_h is also practically constant and does not change with decreasing $<e>$. Similar situations also occur in Frac2–Frac8. However, owing to the randomness of the rough fracture surface, the initial distribution characteristics of the fracture aperture differ between different samples, resulting in different stages of change in the contact rate of the samples; R and Q also change accordingly.

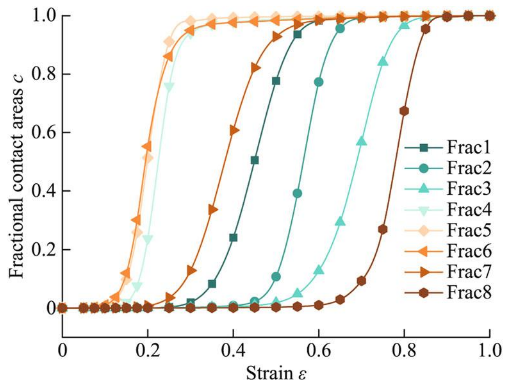

FIGURE 8 Changes in the contact area c of Frac1 to Frac8 with strain.

As shown in Figure 10a, at $\varepsilon = 0.35$, R is 92.28%, which is reduced by 3.75% compared to $R = 96.03\%$ at $\varepsilon = 0.3$, whereas c increased from 0.36% to 1.97%, showing an increase of 1.61%. Similarly, at $\varepsilon = 0.45$, R is 12.8%, which is reduced by 46.03% compared to 58.91% at $\varepsilon = 0.4$, and the corresponding c value increased by 26.40% from 24.13% to 50.53%. Moreover, R is not linearly reduced with c. Different c values may have different effects on the change of R. Therefore, to better describe this change, ΔR

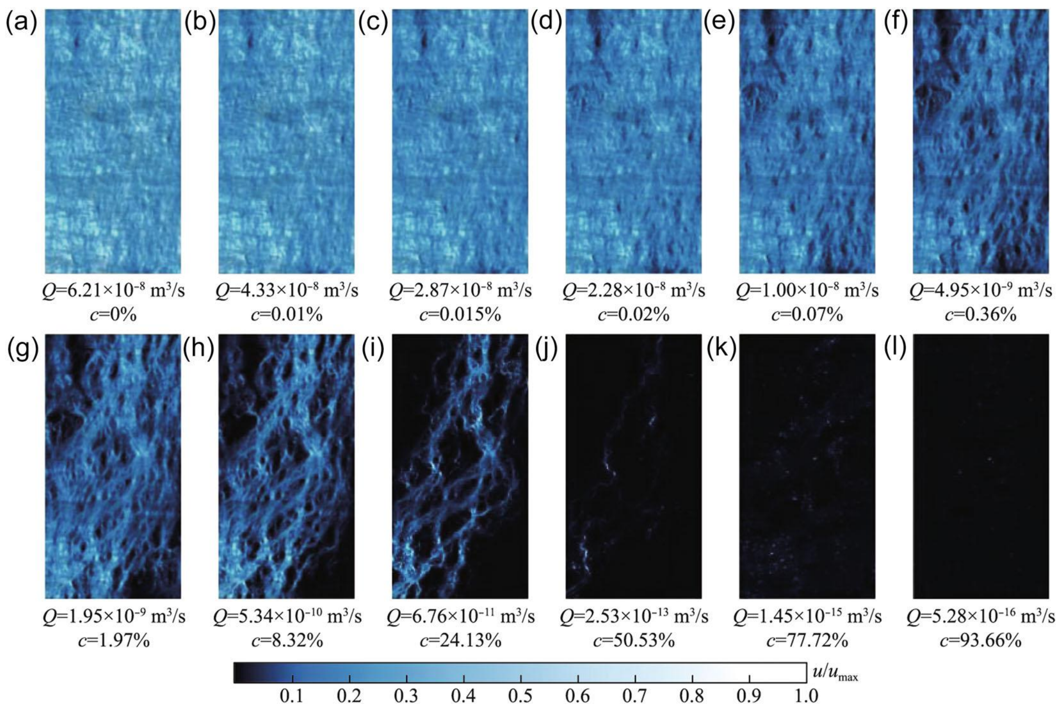

FIGURE 9 Standardized seepage velocity field of Frac1. (a) $\varepsilon = 0$, (b) $\varepsilon = 0.05$, (c) $\varepsilon = 0.10$, (d) $\varepsilon = 0.15$, (e) $\varepsilon = 0.20$, (f) $\varepsilon = 0.25$, (g) $\varepsilon = 0.30$, (h) $\varepsilon = 0.35$, (i) $\varepsilon = 0.40$, (j) $\varepsilon = 0.45$, (k) $\varepsilon = 0.50$, and (l) $\varepsilon = 0.55$.

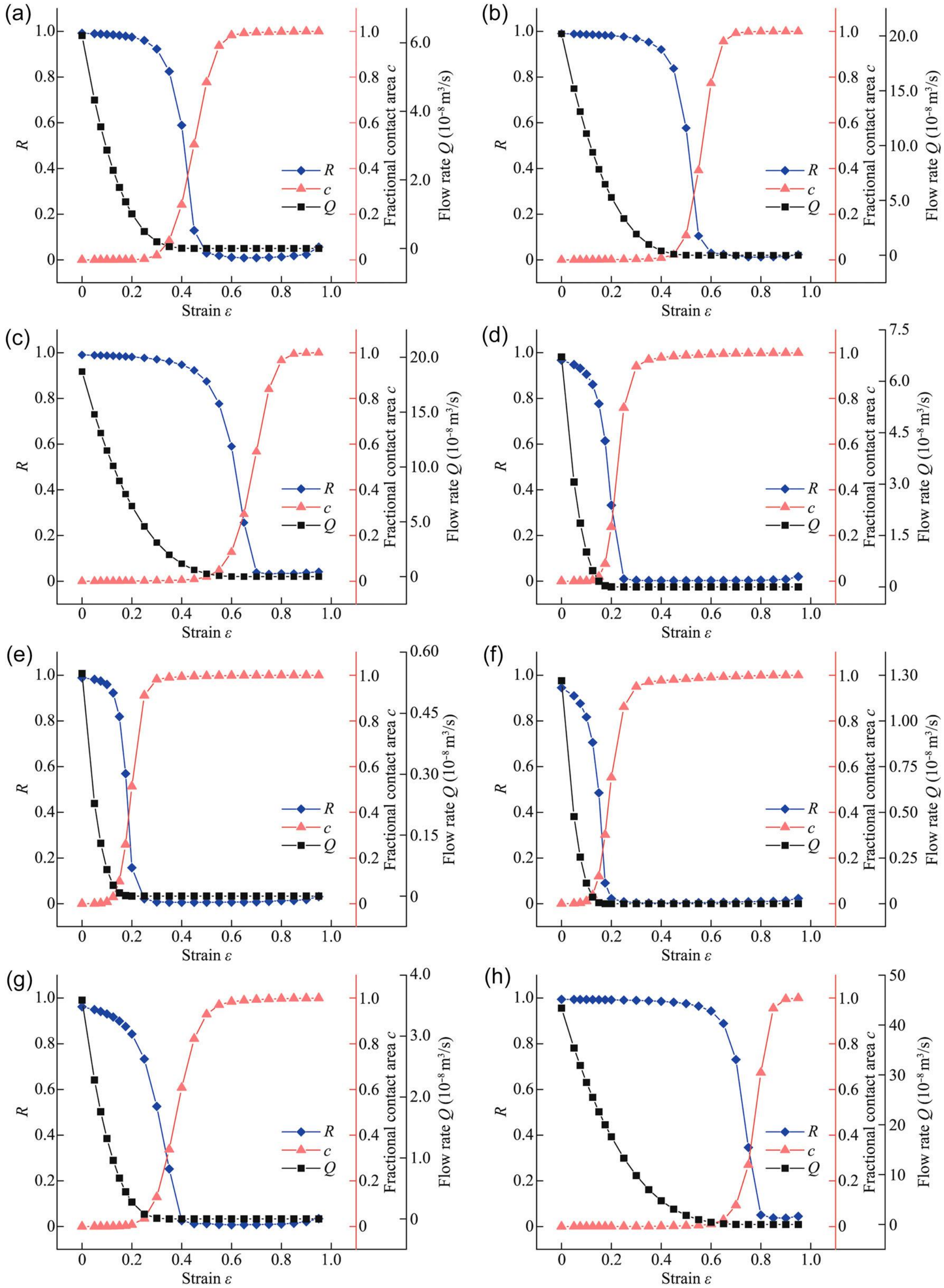

FIGURE 10 Relationship between flow rate, contact area, and R with strain ε. (a) Frac1, (b) Frac2, (c) Frac3, (d) Frac4, (e) Frac5, (f) Frac6, (g) Frac7, and (h) Frac8.

and Δc are proposed. ΔR is defined as the difference between $R_{\varepsilon i\text{-}1}$ and $R_{\varepsilon i}$, expressed as $\Delta R_{\varepsilon i} = R_{\varepsilon i}\text{-}R_{\varepsilon i\text{-}1}$, and Δc is defined as $\Delta c = c_{\varepsilon i}\text{-}c_{\varepsilon i\text{-}1}$. The changes of ΔR and Δc with ε are shown in Figure 11. Figure 11a clearly demonstrates distinct variations between ΔR and Δc of Frac1. Before $\varepsilon = 0.25$, ΔR and Δc exhibit low values with no significant changes. When $0.25 < \varepsilon \leq 0.45$, c ranges from 0.36% to 24.13%. There was a notable change in ΔR, with the reduction exceeding the increase of Δc. This highlights the pronounced inhibitory effect of c on hydraulic fracture aperture e_h. For $0.45 < \varepsilon \leq 0.50$, c ranges from 24.13% to 50.53%. Here, ΔR decreases rapidly, and Δc increases significantly. Although the weakening effect of c on the fracture aperture persists, it decreases noticeably. At the same ε, a higher Δc does not lead to a greater decrease in e_h. For $\varepsilon > 0.5$, although c continues to increase, ΔR almost

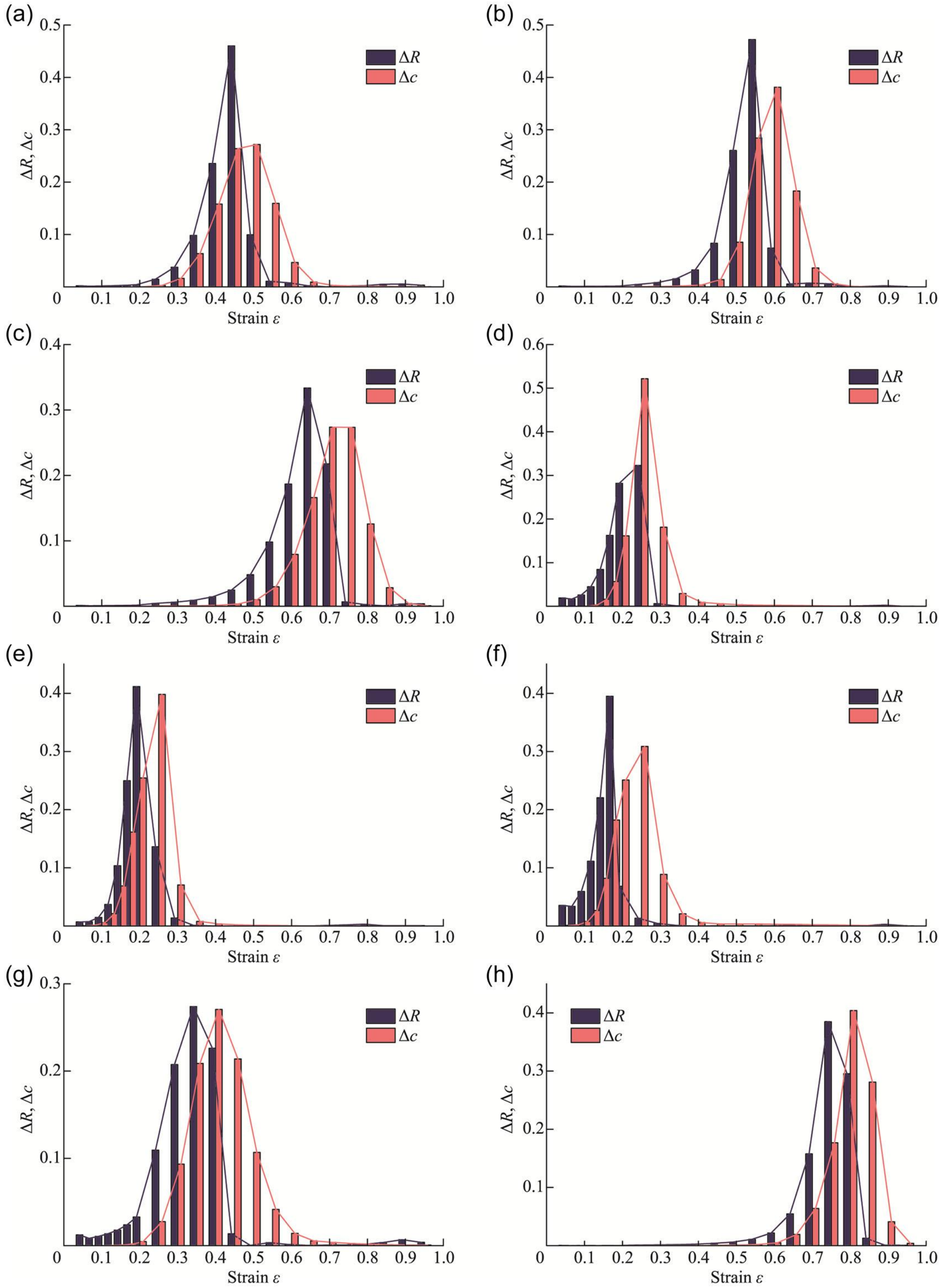

FIGURE 11 Variation of ΔR and Δc with strain ε. (a) Frac1, (b) Frac2, (c) Frac3, (d) Frac4, (e) Frac5, (f) Frac6, (g) Frac7, and (h) Frac8.

ceases to decrease, indicating that when $c > 50.53\%$, the closure of fractures has minimal effect on hydraulic fracture aperture. This phenomenon is primarily attributed to the basic closure of the fracture flow pathways, which prevents further fluid flow within the fractures, as shown in Figure 9.

As observed in Figure 11b–e, the peaks of ΔR always precede the peaks of Δc. Consequently, we separately recorded the strain, ΔR, Δc, and c corresponding to the maximum values of ΔR and Δc for Frac1–Frac8, as shown in Table 2. Owing to the inability to obtain the values of R and c fully and comprehensively under arbitrary strain conditions ε, the results obtained correspond to c values with certain deviations. When ΔR exceeds 1%, it marks the beginning of the effect of c on e_h. For Frac1–Frac8, the range of c corresponding to $\Delta R = 1\%$ is 0.01%–0.77%, with an average of 0.34%. The distribution range of c corresponding to max ΔR is 9.31%–24.13%, with an average c of 14.41%.

This suggests that, under a low normal stress or strain, the influence of c can be ignored. Changes in e_h resulted in changes in the flow rate, which is consistent with the cubic law. Subsequently, the weakening effect of c on e_h gradually increased and then rapidly decreased, during which e_h gradually deviated from the cubic law and eventually fell below the mean fracture aperture. The primary impact of c on e_h weakening is [0.01%, 0.77%] to [9.31%, 24.13%], with an average range from 0.34% to 14.41%. After 14.41%, the influence of c on e_h gradually diminishes because the fractures contain less fluid, and the e_h of the fractures approaches the minimum value. This understanding is crucial for engineering applications. In practical engineering, the relationship between stress and fracture deformation can be established to estimate the contact area of rough fractures. This allowed us to quickly determine the flow direction within the fractured rock masses and select the appropriate flow equations to further describe the permeability characteristics.

3.4 | Interdependence between flow and strain in a mesoscopic fracture under normal stress loading

Using the outlet flow rate of the fracture and injection pressure, the permeability of the fracture can be obtained, and the corresponding fracture hydraulic aperture can be calculated using the cubic law. The relationship between permeability k, hydraulic aperture e_h, and fracture aperture strain ε is shown in Figure 12. Figure 12a shows that the permeability of the fracture decreases with ε, and the decrease rate is gradually weakened. Frac8 maintains its maximum permeability during fracture closure. Conversely, Frac5 exhibits the lowest permeability during this process. The permeability of Frac2 and Frac4 decreases faster than that of the other fractures. This can be attributed to the initial e_m value, which determines the initial permeability. Frac8 had the maximum initial e_m, whereas the initial e_m of Frac5 is the minimum. The distribution range of the initial e_m influenced the rate of decrease e_m. The tight distribution of e_m weakened the permeability of the fracture more rapidly. The later the change in c caused by fracture closure occurs, the slower the decreasing rate of the crack flow with strain.

Figure 12b shows the change of hydraulic aperture e_h with ε_m. This process can be divided into two stages. First, e_h linearly decreases with ε_m; e_h at $\varepsilon_m = 0$ is the initial hydraulic aperture in the natural stress. e_h approaches 0 at different ε_m. Second, when e_h is close to 0, ε_m does not change. This more intuitively indicates that although fracture mesoscopic deformation still occurs, e_h no longer changes, and the fracture closure and fracture mesoscopic deformation do not influence the fracture seepage.

Hydraulic aperture strain ε_h represents the ratio of hydraulic aperture to the initial hydraulic aperture, which can be calculated using Equation (4). Ma et al. (2023) have proven that the ε_h has a linear relation with

TABLE 2 Specific parameters at different ΔR.

No.	$\Delta R = 1\%$ ε	ΔR	c (%)	MaxΔR ε	ΔR	c (%)
Frac1	0.25	0.014666375	0.36	0.45	0.460278095	24.13
Frac2	0.35	0.015465878	0.46	0.55	0.472220431	10.75
Frac3	0.40	0.005409239	0.50	0.65	0.333450903	12.75
Frac4	0.05	0.000523308	0.01	0.25	0.322775660	23.77
Frac5	0.10	0.011412336	0.77	0.2	0.249689442	9.79
Frac6	0.05	0.049698650	0.10	0.2	0.394906993	11.96
Frac7	0.10	0.020612843	0.02	0.35	0.273941180	12.86
Frac8	0.55	0.034171534	0.51	0.75	0.384794919	9.31

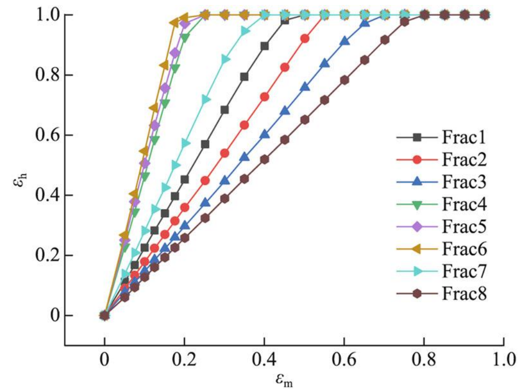

FIGURE 13 Relationship between the strains in terms of the mechanical/hydraulic aperture ε_m–ε_h.

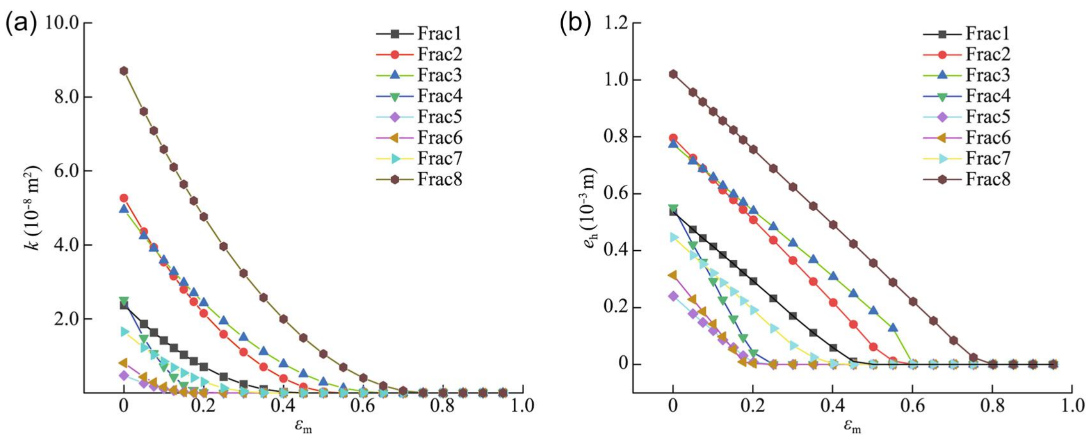

FIGURE 12 Relationship between the intrinsic permeability, hydraulic aperture, and strain. (a) The permeability of fracture decrease with ε; (b) the change of hydraulic aperture e_h with ε_m.

ε_m, and various fitting equations were obtained to predict the change of hydraulic aperture. In this study, the relationship between ε_h and ε_m of Frac1–Frac8 was plotted in Figure 13. It is evident that at first, ε_h linearly increases with ε_m. When ε_h is 1, the hydraulic fracture opening reaches the minimum value; then, ε_h remains 1 and no longer decreases with fracture closure.

4 | CONCLUSION

To further investigate the influence of mesoscopic deformation of fracture on the contact area and hydraulic aperture of fracture seepage, natural-splitting fractures were generated and digitally reproduced using a three-dimensional scanner, and numerical simulations of fluid flow between rough surfaces were conducted. Based on these, the influence of fracture closure on the contact area and mechanical aperture, the interdependence between the contact area and fluid flow behavior, and the relationship between the fluid flow and strain in a single mesoscopic fracture were investigated, respectively.

1. The initial mechanical apertures of fractures can be described using a normal distribution. As the fracture closes, the change in contact area with strain exhibits three stages, transitioning from slow variation to rapid growth, before eventually stabilizing. The contact area variation process varies for different rough fractures and is primarily influenced by the mean and maximum fracture apertures.
2. The seepage velocity field changed significantly with fracture closure. When the normal displacement was low, it did not significantly alter the flow path; however, the flow rate exhibited an evident decrease. With an increase in displacement, the flow path changed significantly, and the velocity field exhibited preferential flow. Subsequently, the preferential flow disappeared, and the flow path of the fluid was closed.
3. The weakening effect of c on e_h varied with fracture closure. At a low strain, c has a weak effect on e_h, at which e_h is identical to e_m. With an increase in c, the flow path gradually closes, e_h gradually deviates from the cubic law, and e_m cannot predict the hydraulic characterization. Subsequently, the effect of c gradually diminishes, and e_h approaches its minimum value and does not change with e_m.

The major weakness of this study, however, is that we focus on the laminar flow for incompressible and steady Newtonian fluids. The turbulence, another possible cause of deviations from the cubic law, is not to be considered. At low Reynolds numbers, viscous forces are strong enough to damp out any perturbations from the unidirectional, laminar flow field, whereas at sufficiently high velocities, small perturbations to the laminar flow field will tend to grow in an unstable manner. These limitations may potentially lead to discrepancies between the simulated and actual behaviors at high velocities. In future research endeavors, we plan to conduct the nonlinear flow considering the fracture aperture and contact area, improving geological engineering applications.

AUTHOR CONTRIBUTIONS

Chenghao Han: Project administration; investigation; funding acquisition; writing—original draft; writing—review draft; writing—review and editing. **Yuanlin Bai:** Data curation; visualization; validation; writing—original draft. **Weijie Zhang:** Supervision; formal analysis. **Jicheng Zhang:** Methodology; formal analysis; writing—original draft; resources; writing—review and editing; **Hongfa Ma:** Writing—original draft; software. **Weiye Li:** Data curation; visualization; conceptualization.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Data that support the findings of this study are available from the corresponding author upon reasonable request.

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